

MATH 261, Homework Set 3

1. Match each function to one of the four gradient vector fields graphed below.

a) $f(x, y) = x^2 + y^2$

b) $f(x, y) = x^2 - y^2$

c) $f(x, y) = y^2 - x^2$

d) $f(x, y) = xy$

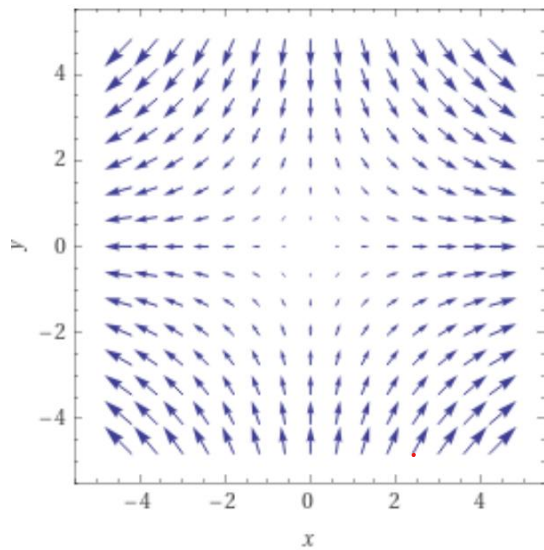


Figure 1

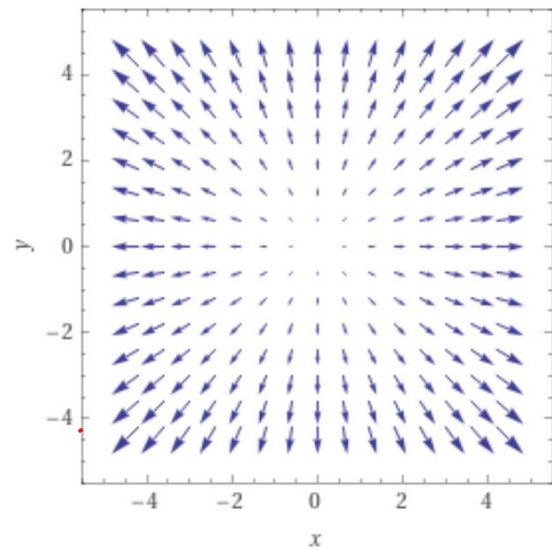


Figure 2

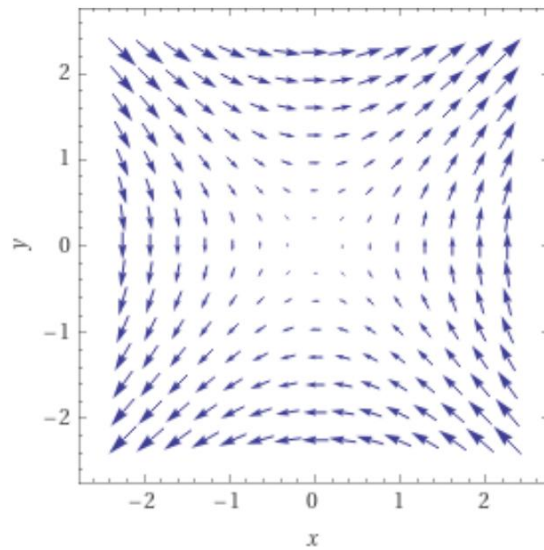


Figure 3

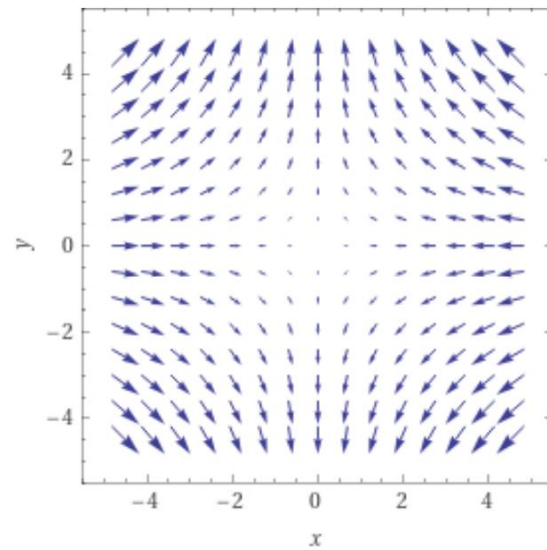


Figure 4

2. Let $f(x, y) = x^3y + y^2x^2 + x^4$. Compute a formula for the two dimensional gradient vector, $\nabla f(x, y)$.
3. Let $f(x, y, z) = x^2 \sin(1 + y^2) + z^3$. Compute a formula for the three dimensional gradient vector, $\nabla f(x, y, z)$.
4. Let $f(x, y) = y^2 \sin(xy)$. In which direction is this function increasing the fastest at the point $(0, 2)$? In which direction is it decreasing the fastest?
5. Let $f(x, y) = ye^x + x^2$. In which direction is this function increasing the fastest at the point $(0, 0)$? In which direction is it decreasing the fastest?
6. Let $f(x, y) = y^4 - 4x^2 + 5$. In which direction is this function increasing the fastest at the point $(1, 1)$? In which direction is it decreasing the fastest?
7. Let $f(x, y, z) = x^2y^2 + z^4$. In which direction is this function increasing the fastest at the point $(1, 1, -1)$? In which direction is it decreasing the fastest?
8. Let $f(x, y, z) = xy^2z^4$. In which direction is this function increasing the fastest at the point $(0, -2, 1)$? In which direction is it decreasing the fastest?
9. Compute $D_{\mathbf{u}}f(x, y)$, where $\mathbf{u}$ points in the direction of $\langle 1, 1 \rangle$ and $f(x, y) = xy^2$, at the point $(2, 2)$.
10. Compute $D_{\mathbf{u}}f(x, y)$, where $\mathbf{u}$ points in the direction of $\langle 0, -1 \rangle$ and $f(x, y) = xy^2$, at the point $(2, 2)$.
11. Suppose that $D_{\mathbf{u}}f(x, y, z)$ is large and positive at $(0, 0, 0)$, for a unit vector $\mathbf{u}$. What does this tell about $f(x, y, z)$? Suppose that $D_{\mathbf{u}}f(x, y, z)$ was zero at the origin instead – what would that tell you?
12. Suppose $\nabla f = \langle 1, 2, -1 \rangle$ at $(0, 0, 0)$ for a function $f(x, y, z)$. What is $D_{\mathbf{u}}f(0, 0, 0)$, if $\mathbf{u} = \langle 0, 1, 3 \rangle$? Is the function increasing or decreasing?
13. Compute the function $D_{\mathbf{u}}f(x, y, z)$, where $f(x, y, z) = yze^x + 1$ and $\mathbf{u}$ points in the direction of $1\mathbf{i} + 0\mathbf{j} + 1\mathbf{k}$.
14. Compute the function $D_{\mathbf{u}}f(x, y, z)$, where $f(x, y, z) = xy^2 + ye^z$ and $\mathbf{u}$ points in the direction of $1\mathbf{i} + 0\mathbf{j} + 2\mathbf{k}$.

15. Compute the function $D_{\mathbf{u}}f(x, y, z)$, where $f(x, y, z) = \frac{y}{x^2+1} + z^8$ and $\mathbf{u}$ points in the direction $1\mathbf{i} - 1\mathbf{j} + 0\mathbf{k}$.
16. Explain, geometrically, why f_x and f_y must both be zero at local maximum or local minimum of a smooth function $f(x, y)$.
17. Show that $(0, 0)$ is a critical point of the function $f(x, y) = x^2 + 3y^2 + x^3y^3$. Use the second partial derivative test to classify this as a local maximum, local minimum or neither.
18. Show that $(1, 1)$ is a critical point of the function $f(x, y) = x^2 + y^2 - 2x - 2y$. Use the second partial derivative test to classify this as a local maximum, local minimum or a saddle point.
19. Show that $(1, 3)$ is a critical point of the function $f(x, y) = 6y + 2x - x^2 - y^2 + 10$. Use the second partial derivative test to classify this as a local maximum, local minimum or a saddle point.
20. Show that $(1, 1)$ is a critical point of the function $f(x, y) = x^2 - 2x - y^2 + 2y + 18$. Use the second partial derivative test to classify this as a local maximum, local minimum or a saddle point.
21. Show that $(0, 0)$ is a critical point of the function $f(x, y) = x^3y + xy$. Use the second partial derivative test to classify this as a local maximum, local minimum or a saddle point.
22. Find a vector normal to the surface $x^2 + y^2 - z^2 = 6$ at the point $(3, 1, 2)$.
23. Find a vector normal to the surface $3z^3 + x^2y - y^2x = 1$ at the point $(1, -1, 1)$.
24. Find an equation of the tangent plane to the surface $x^2 + 3y^2 + 4z^2 = 20$ at the point $(2, 2, 1)$.
25. Find an equation of the tangent plane to the surface $xz + 2x^2y + y^2z^3 = 11$ at the point $(2, 1, 1)$.